Overview of methods for spectral estimation

WS Stationary signal (or "quasi"-stationary)

- Non-parametric ("black-box")
  - DFT-based
    - DFT of one block
      - ***Periodogram***
      - ***(Modified periodogram)***
    - DFT averaging
      - ***Bartlett averaging (non-overlapping blocks)***
      - ***Welch averaring (overlapping blocks)***
  - Correlation-based
      - ***Blackman-Tukey***

- Parametric (model-based, "grey-box")
  - LTID-system based signal model
    - All-pole (IIR) LTID-system
      - ***AR (autoregressive) spectral estimation***
    - All-zeros (FIR) LTID-system
      - ***MA (moving average) spectral estimation***
    - General (IIR) LTID-system
      - ***ARMA (autoregressive-moving average) spectral estimation***

- Eigenvalue-based ("frequency estimation")
  - The eigenvalue method
  - Pisarenko decomposition
  - The MUSIC-algorithm

Non-stationary signals (time/frequency analysis)

- ***STFT, Short-time FT, Sliding window FT, Spectrogram***
- *Wavelet-analysis*
- Discrete Cosine Transform
- Walsh Transform